

RW False-Positive Audit — B3 condition

91 findings to review · 29 papers

What you're doing

SPOT annotates exactly *one* error per paper — the one that caused the retraction or correction. But papers usually have more than one severe issue.

The aggregate metric "RW-precision = matched / total" counts every RW finding outside the single SPOT annotation as a false positive. Some of those are **genuine severe errors that SPOT just didn't annotate**. This audit identifies them.

For each finding, choose one verdict:

- **VALID SEVERE** — genuine severe methodological error that, if confirmed, would warrant retraction. The arbiter's RW classification is correct; SPOT just didn't annotate this one.
- **RELATED** — the finding identifies a real issue, but the severity rating is too high. Should have been MAJOR-REVISION. Use this when you're uncertain — it's the conservative choice.
- **FALSE ALARM** — not a genuine severe error, or the model misread the paper.

Calibration guidance

- **VALID** means: *"if true, this paper should be retracted, independent of SPOT's annotation."* Be strict.
- If you can't verify the technical claim (math proof, materials chemistry, specific theorem), mark **RELATED** rather than guessing.
- The SPOT-annotated error is shown above each finding for context only — it is NOT the standard. The standard is whether *this specific finding* describes a retraction-worthy error in the paper.

How to record your verdict

This PDF is read-only. After reading each finding here, open the matching CSV file in a spreadsheet (Excel, Numbers, or VS Code) and fill in the `human_verdict` and `notes` columns:

```
analysis/rw_false_positive_audit/audit_sheet_b3.csv
```

Use exactly **VALID**, **RELATED**, or **FALSE** in the `human_verdict` column.

Time estimate

2–5 minutes per finding. Full audit ≈ 3–7 hours. You can stop and resume — the CSV holds your progress.

What we're looking for

Whether GD has a higher rate of VALID findings (among its unmatched RW pool) than B3 or B3++. If yes, the steelman exchange is selecting for real severe errors at a higher rate than alternative reflection protocols — and the original RW-precision metric was understating GD's advantage.

If the rates are similar, the calibration advantage is exactly as the aggregate metric showed (2× directional, not significant at this n). If GD's rate is lower, the aggregate metric was overstating GD. The honest answer is whatever the data says.

10.1016_j.jpcc.2024.112058 — Facile synthesis of SnSe,ÀMnTe nanocomposite as a promising electrode for supercapacitor applications

- **Paper category:** Materials Science
- **SPOT error category:** Data Inconsistency (figure-text) (severity tier: errata)
- **SPOT annotation (the one annotated error):** The error is that the EDX spectrum in Figure 2d was incorrectly labeled, showing inconsistencies such as a flat background and missing expected iron peaks, which led to incorrect assignment of elements to peaks.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

Energy density and power density are computed from three-electrode half-cell data and presented as supercapacitor device performance metrics. The specific energy formula used ($SE = Cs \times \Delta V^2 / 7.2$) appears to omit the factor of $1/2$, yielding ~ 61 Wh/kg instead of the correct ~ 44 Wh/kg (using $Cs = 1502$ F/g and $\Delta V = 0.65$ V with the standard $E = 0.5 \cdot Cs \cdot \Delta V^2 / 3.6$). This inflated headline result is the paper's central performance claim.

- **Arbiter justification for RW classification:**

Both reviewers flagged this. The combination of (a) using a three-electrode potential window to compute device-level energy/power density and (b) an incorrect formula that over-estimates SE by roughly 40% means the paper's primary quantitative conclusion—outstanding energy density for supercapacitor application—is fundamentally unsupported by the data as presented. Recalculation with the correct formula and appropriate device-level methodology would yield substantially different numbers, invalidating the comparison tables and the paper's positioning against the literature.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

10.1016_j.scenv.2024.100142 — Extraction and characterization of biocompatible hydroxyapatite (Hap) from red big eye fish bone: Potential for biomedical applications and reducing biowastes

- **Paper category:** Environmental Science
- **SPOT error category:** Data Inconsistency (figure-text) (severity tier: errata)
- **SPOT annotation (the one annotated error):** Figures 1 and 2 are supposed to display FT-IR and XRD spectra, but instead show SEM images, indicating the figure images and their legends are mismatched and do not represent the described data.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

The XRD discussion contains extended text about 'Fe-Al-based mesoporous metal oxides,' 'BMO,' and 'Cu K α peaks in BMO' that are entirely irrelevant to a fish-bone hydroxyapatite study, indicating text recycling or contamination from an unrelated manuscript.

- **Arbiter justification for RW classification:**

The presence of clearly unrelated scientific content in the results/discussion section indicates either plagiarism, text fabrication, or such severe carelessness that the integrity of the entire analytical interpretation is called into question. If confirmed as recycled text, readers cannot trust that the XRD interpretation actually corresponds to the samples studied. This undermines a core characterization pillar of the paper.

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-

10.1038_s41598-024-56334-5 — The impact understanding of exosome therapy in COVID-19 and preparations for the future approaches in dealing with infectious diseases and inflammation

- **Paper category:** Multidisciplinary
- **SPOT error category:** Data Inconsistency (figure-text) (severity tier: errata)
- **SPOT annotation (the one annotated error):** The error is that the figure shows a micrograph from a TESCAN MIRA 3 device, whereas the text in the Materials and Methods section incorrectly describes the instrument used as a Hitachi S-4160 and Zeiss EM10C, resulting in a discrepancy between the described methods and the actual data presented.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

Fundamental mismatch between study design and claimed conclusions: The paper claims to assess the efficacy and safety of exosome therapy in COVID-19 patients, but only performs an in vitro PBMC stimulation experiment on 5 patients and 5 controls. No patients received exosomes. The title, abstract, and conclusions misleadingly imply a clinical intervention.

- **Arbiter justification for RW classification:**

Both reviewers independently identify this as the core fatal flaw. The paper's central conclusions about therapeutic safety and efficacy in COVID-19 patients are not supported by in vitro data. The data at most support a limited observation about exosome-mediated cytokine modulation in cultured PBMCs. The disconnect between the experimental design and the headline claims is not a matter of reframing—it fundamentally invalidates the paper's stated conclusions. A complete rewrite reframing the study as a preliminary in vitro observation would constitute a different paper.

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- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

Unsupported safety claims: The paper explicitly states findings 'underscore the potential safety and efficacy' of exosomes as a therapeutic strategy, but no toxicity, safety, or tolerability evaluation was performed in patients or animals. An MTT viability assay on PBMCs does not constitute evidence of clinical safety.

- **Arbiter justification for RW classification:**

This is a specific and egregious instance of the design-conclusion mismatch. Safety in humans cannot be inferred from PBMC viability data. Since safety is explicitly part of the paper's stated conclusion, this claim is fundamentally unsupported. This compounds the first finding and reinforces the retraction-worthy assessment.

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10.1038_s41598-024-73921-8 — Assessing the effect of therapeutic level of oxytetracycline dihydrate on pharmacokinetics and biosafety in *Oncorhynchus mykiss* (Walbaum, 1792)

- **Paper category:** Multidisciplinary
- **SPOT error category:** Statistical reporting (severity tier: errata)
- **SPOT annotation (the one annotated error):** The error is that negative values are reported for pharmacokinetic parameters such as AUC, elimination rates, and LOQ in Table 1, which are nonsensical since these parameters should always be positive based on their definitions in pharmacokinetics and analytical chemistry.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

Impossible analytical validation values: negative LOD/LOQ (e.g., LOD = -587.51 µg/kg, LOQ = -1780.34 µg/kg in kidney) and R^2 values outside the valid 0–1 range (e.g., muscle R^2 = 6.96, skin R^2 = 0.002). The authors' claim that negative LOD/LOQ values 'affirm method impartiality' is scientifically incorrect.

- **Arbiter justification for RW classification:**

Negative concentration limits are physically meaningless, and $R^2 > 1$ is mathematically impossible. These are not ambiguous analytical choices — they indicate that either the calibration/validation was fundamentally flawed or the values are gross reporting errors. Since all tissue concentration data depend on validated analytical methods, the entire quantitative dataset (PK concentrations, residue levels, withdrawal times) is rendered unreliable. Both reviewers flagged this independently with the highest severity.

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- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

Pharmacokinetic parameters are internally contradictory and mathematically inconsistent. Elimination half-lives do not match reported elimination rate constants via $t_{1/2} = \ln(2)/k$ (e.g., $k = 0.01 \text{ h}^{-1}$ should give $t_{1/2} \approx 69 \text{ h}$, not 0.94 h). Reported half-lives include physiologically implausible values (plasma $t_{1/2} = 3368.24 \text{ h} \approx 140 \text{ days}$). The same tissue (gill) appears with contradictory half-life values (0.94 h , 198.48 h , 19.39 h) in different sections. $T_{\max}/C_{\max}$ values contradict between abstract, tables, and text.

- **Arbiter justification for RW classification:**

The pharmacokinetic characterization is the manuscript's primary contribution. When half-lives, rate constants, $T_{\max}$, and $C_{\max}$ values are mutually contradictory and mathematically inconsistent, no reliable PK profile can be extracted. These are not borderline discrepancies but order-of-magnitude contradictions that cannot be resolved by the reader. Both reviewers identified this as retraction-worthy.

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- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

Fundamental misapplication of non-compartmental analysis (NCA). The manuscript claims NCA but reports absorption rate constants (K_a), absorption half-lives ($t_{1/2a}$), distribution phases, and biphasic compartmental interpretations. Standard NCA from oral data cannot directly estimate K_a without IV reference data or deconvolution. The reported $t_{1/2a}$ and $t_{1/2b}$ parameters imply a two-compartment model that was never formally fitted or justified.

- **Arbiter justification for RW classification:**

The entire PK parameter set is derived from a methodology that is incorrectly described and apparently incorrectly applied. Parameters like K_a from NCA of oral-only data are not valid without additional modeling assumptions that are neither stated nor justified. This means the absorption, distribution, and elimination characterization — the core of the paper — is built on an invalid analytical framework. Both reviewers flagged this.

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- **B3 finding (rated RETRACTION-WORTHY, finding #3):**
Location:

The reported AUC trapezoidal formula omits the required 1/2 factor: manuscript gives $\Sigma(C_i + C_{i+1}) \times (\Delta t)$ instead of the correct $\Sigma((C_i + C_{i+1})/2) \times (\Delta t)$, which would double all AUC values.

- **Arbiter justification for RW classification:**

AUC is a central pharmacokinetic exposure metric used for tissue distribution comparisons and regulatory interpretations. If the stated formula was actually used, all AUC values are doubled and exposure estimates are systematically wrong. If the formula is merely misreported, it reflects a level of carelessness that compounds the other errors. Given the other mathematical inconsistencies, it is plausible the formula error is real.

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- **B3 finding (rated RETRACTION-WORTHY, finding #4):**
Location:

Sampling/replication structure is unclear and likely pseudoreplicated. Three fish per tank were pooled into one tissue sample; tanks may have been in triplicate, but statistical analyses report means $\pm$ SD without clarifying the unit of analysis. For PK, each time point may have only $n=1$ pooled value, making inferential statistics on PK parameters impossible.

- **Arbiter justification for RW classification:**

If pooled samples were treated as independent fish-level replicates, all variance estimates and p-values are invalid. For the PK study, $n=1$ pooled samples per time point mean no inter-individual variability can be estimated, which is essential for reliable PK parameter estimation. While pooled designs exist in destructive sampling studies, the manuscript's statistical claims go well beyond what such a design supports. The severity is high because it affects both the PK and biosafety inferential conclusions, though some uncertainty remains about the exact replication structure.

- **Human verdict** (mark one):
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 - **Notes:** _____
-

10.1038_s41598-025-85894-3 — Photocatalytic removal of textile wastewater-originated methylene blue and malachite green dyes using spent black tea extract-coated silver nanoparticles

- **Paper category:** Multidisciplinary
- **SPOT error category:** Data inconsistency (severity tier: errata)
- **SPOT annotation (the one annotated error):** The error in Fig. 4 is that the UV-Vis spectrum for the "chemically synthesized Ag nanoparticles" displays an artifact—a sharp, anomalous peak at ~260 nm—due to unfiltered junk data not being removed before plotting, and the horizontal axis label is also misspelled.

- **B3 finding (rated RETRACTION-WORTHY, finding #0):**

Location:

Missing essential controls make it impossible to attribute observed absorbance decreases to photocatalysis. No dark control (to quantify adsorption), no light-only control (to quantify photolysis), no spent tea extract-only control, and no proper baseline were included. The entire mechanistic attribution of 'photocatalytic degradation' is unsupported.

- **Arbiter justification for RW classification:**

Both reviewers identify this as a fundamental flaw. Without these controls, the observed absorbance decrease could be entirely due to adsorption, photolysis, sedimentation, or spectral interference from nanoparticles. The paper's central conclusion—that SBT-AgNPs photocatalytically degrade dyes—cannot be sustained from the presented data. This is not a matter of improving the study; the core inference is broken.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

10.1038_s41598-025-85894-3 — Photocatalytic removal of textile wastewater-originated methylene blue and malachite green dyes using spent black tea extract-coated silver nanoparticles

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- **SPOT annotation (the one annotated error):** The error in Fig. 4 is that the UV-Vis spectrum for the "chemically synthesized Ag nanoparticles" displays an artifact—a sharp, anomalous peak at ~260 nm—due to unfiltered junk data not being removed before plotting, and the horizontal axis label is also misspelled.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

Dye identification in real textile wastewater is based solely on matching UV-Vis absorption peaks/humps, with no chromatographic confirmation, spike-recovery experiments, or matrix correction. In a complex effluent, multiple absorbing species can produce overlapping peaks that mimic target dyes.

- **Arbiter justification for RW classification:**

Reviewer A identifies this as retraction-worthy, and Reviewer B raises the same concern through the wastewater characterization finding. If the dyes are not actually present or are not the species being measured, the entire applied wastewater claim collapses. Combined with the lack of controls, this invalidates the paper's principal application claim.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
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- **B3 finding (rated RETRACTION-WORTHY, finding #2):**

Location:

Quantification of dye removal in wastewater using raw absorbance changes at selected wavelengths without matrix-matched calibration, baseline correction, deconvolution, or accounting for nanoparticle scattering/adsorption. Absorbance decrease $\neq$ photocatalytic degradation.

- **Arbiter justification for RW classification:**

Both reviewers note that the quantitative removal percentages are derived without validated calibration in the wastewater matrix. The presence of nanoparticles in the measured solution introduces scattering artifacts. Without separating these confounds, the reported degradation percentages are meaningless, directly undermining the paper's quantitative conclusions.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
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 - **Notes:** _____
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10.1038_s41598-025-85894-3 — Photocatalytic removal of textile wastewater-originated methylene blue and malachite green dyes using spent black tea extract-coated silver nanoparticles

- **Paper category:** Multidisciplinary
- **SPOT error category:** Data inconsistency (severity tier: errata)
- **SPOT annotation (the one annotated error):** The error in Fig. 4 is that the UV-Vis spectrum for the "chemically synthesized Ag nanoparticles" displays an artifact—a sharp, anomalous peak at ~260 nm—due to unfiltered junk data not being removed before plotting, and the horizontal axis label is also misspelled.

- **B3 finding (rated RETRACTION-WORTHY, finding #3):**

Location:

Impossible or internally inconsistent dye concentrations reported in the methods. Malachite green is described at 100 mg/ml (100,000 ppm), which far exceeds solubility and would saturate any UV-Vis measurement. Methylene blue concentrations are reported inconsistently as masses and concentrations. These errors make the degradation calculations physically implausible or unreproducible.

- **Arbiter justification for RW classification:**

Reviewer B identifies this as retraction-worthy. If the concentrations are as stated, the UV-Vis data are physically impossible; if they are reporting errors, all derived degradation percentages are based on incorrect inputs and are meaningless. Either way, the quantitative core of the paper is invalidated. Reviewer A corroborates this through findings about unit inconsistencies and missing calibration.

- **Human verdict** (mark one):
 - ☐ **VALID SEVERE** — genuine error, SPOT missed it
 - ☐ **RELATED** — real issue but severity overstated
 - ☐ **FALSE ALARM** — not a genuine severe error
 - **Notes:** _____
-

10.1186_s12964-024-01868-4 — Illuminating the dark kinome: utilizing multiplex peptide activity arrays to functionally annotate understudied kinases

- **Paper category:** Biology
- **SPOT error category:** Reagent identity (severity tier: errata)
- **SPOT annotation (the one annotated error):** The error is that the recombinant protein fragments used for at least three kinases do not include their respective kinase domains, meaning the reagents do not have intrinsic kinase activity and the experimental results do not reflect actual kinase function for those proteins.

- **B3 finding (rated RETRACTION-WORTHY, finding #0):**

Location:

Schizophrenia/control and sex-specific comparisons are based on single pooled samples per group with only technical triplicates, providing no biological replication for population-level inference.

- **Arbiter justification for RW classification:**

Both reviewers independently identify this as the most critical flaw. A single pooled lysate per condition (male control, male SCZ, female control, female SCZ) with technical triplicates cannot estimate between-subject variance. All statistical comparisons, heatmaps, upstream kinase analyses, and sex-specific claims derived from these pooled samples are pseudoreplicated. The disease-association conclusions—a central pillar of the paper—are fundamentally unsupported by the experimental design. This is not a matter of being underpowered; the design structurally cannot answer the questions posed.

- **Human verdict** (mark one):
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 - ☐ RELATED — real issue but severity overstated
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10.5539_jsd.v17n6p137 — The Scientific Case Against Net Zero: Falsifying the Greenhouse Gas Hypothesis

- **Paper category:** Environmental Science
- **SPOT error category:** Equation / proof (severity tier: errata)
- **SPOT annotation (the one annotated error):** The author used the base-10 logarithm (log10) instead of the natural logarithm (ln, neperian logarithm) in the IPCC radiative forcing equation, resulting in incorrect temperature change calculations.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

No reproducible review methodology: no search strategy, databases, inclusion/exclusion criteria, quality assessment, or synthesis protocol is provided for what is presented as an evidence-based literature review.

- **Arbiter justification for RW classification:**

The manuscript's entire contribution depends on its review-based inference. Without a valid, reproducible methodology, the central conclusions cannot be said to arise from a scientific process. Both reviewers identify this as a foundational failure.

- **Human verdict** (mark one):
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- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

The manuscript claims the greenhouse effect is 'fictitious' and that planetary surface temperatures are explained by the ideal gas law alone, without radiative transfer considerations.

- **Arbiter justification for RW classification:**

This contradicts over a century of experimentally verified atmospheric physics. The ideal gas law relates state variables and cannot determine a planet's energy balance or equilibrium temperature without radiative constraints. This error is central to the paper's thesis and invalidates its core physical argument.

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- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

The manuscript derives or endorses a hard upper limit on CO₂-driven warming (~0.5–0.81°C) by misusing the logarithmic forcing relationship and ignoring climate feedbacks.

- **Arbiter justification for RW classification:**

Logarithmic forcing is the standard formulation and does not imply negligible warming; it is consistent with substantial warming per CO₂ doubling when feedbacks are included. The paper's central quantitative conclusion about climate sensitivity is built on this misinterpretation. If corrected, the main policy conclusion collapses.

- **Human verdict** (mark one):
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- **SPOT annotation (the one annotated error):** The author used the base-10 logarithm ($\log_{10}$) instead of the natural logarithm ($\ln$, neperian logarithm) in the IPCC radiative forcing equation, resulting in incorrect temperature change calculations.
- **B3 finding (rated RETRACTION-WORTHY, finding #3):**
Location:

The manuscript states that the increase in global temperature from 1880–2000 is 'statistically indistinguishable from 0°K,' contradicting all major observational temperature datasets.

- **Arbiter justification for RW classification:**

The ~0.8°C warming over this period is documented with clear statistical significance by NASA GISS, NOAA, HadCRUT, and Berkeley Earth. This claim is empirically false and directly supports the paper's conclusion that warming is not occurring. Both reviewers flag this.

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- **B3 finding (rated RETRACTION-WORTHY, finding #4):**
Location:

The manuscript conflates CO_2 residence time with adjustment time, and calculates that anthropogenic CO_2 is only ~4.3% of atmospheric CO_2 by comparing human emissions to gross natural fluxes, ignoring net imbalance and stock-flow accounting.

- **Arbiter justification for RW classification:**

This is a well-known carbon-cycle error. Natural gross fluxes are largely balanced; the anthropogenic perturbation is the net addition causing the observed CO_2 rise. The paper's claim that human emissions are negligible depends entirely on this confusion. Both reviewers identify this error.

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- **Paper category:** Environmental Science
- **SPOT error category:** Equation / proof (severity tier: errata)
- **SPOT annotation (the one annotated error):** The author used the base-10 logarithm ($\log_{10}$) instead of the natural logarithm ($\ln$, neperian logarithm) in the IPCC radiative forcing equation, resulting in incorrect temperature change calculations.
- **B3 finding (rated RETRACTION-WORTHY, finding #5):**
Location:

The source base is dominated by advocacy organizations, blogs, opinion books, think-tank reports, YouTube videos, and preprints, treated as equivalent to peer-reviewed primary literature.

- **Arbiter justification for RW classification:**

For a review article making sweeping scientific claims that reject mainstream physics, the evidentiary base must be robust. Systematic reliance on non-scholarly and partisan sources to reject established science fatally compromises the review's validity. Both reviewers note this, though Reviewer B initially classified it as minor before acknowledging the scale makes it more serious.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

10.5539_jsd.v17n6p137 — The Scientific Case Against Net Zero: Falsifying the Greenhouse Gas Hypothesis

- **Paper category:** Environmental Science
- **SPOT error category:** Equation / proof (severity tier: errata)
- **SPOT annotation (the one annotated error):** The author used the base-10 logarithm (log10) instead of the natural logarithm (ln, neperian logarithm) in the IPCC radiative forcing equation, resulting in incorrect temperature change calculations.
- **B3 finding (rated RETRACTION-WORTHY, finding #6):**
Location:

Extreme selection bias: the review omits the principal lines of evidence bearing on its thesis—observed energy imbalance, attribution studies, paleoclimate constraints, fingerprinting, ocean heat content, and multiple independent temperature datasets.

- **Arbiter justification for RW classification:**

A review that selectively samples only contrarian material while claiming to adjudicate an entire field cannot support its conclusions. The omission of the strongest contrary evidence is a fatal synthesis failure, not merely an oversight.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
- **Notes:** _____

10.5539_jsd.v17n6p137 — The Scientific Case Against Net Zero: Falsifying the Greenhouse Gas Hypothesis

- **Paper category:** Environmental Science
- **SPOT error category:** Equation / proof (severity tier: errata)
- **SPOT annotation (the one annotated error):** The author used the base-10 logarithm ($\log_{10}$) instead of the natural logarithm ($\ln$, neperian logarithm) in the IPCC radiative forcing equation, resulting in incorrect temperature change calculations.
- **B3 finding (rated RETRACTION-WORTHY, finding #7):**
Location:

The manuscript confuses radiative transfer with thermal conductivity, implying the greenhouse effect should manifest as a bulk thermal conductivity anomaly in laboratory gases.

- **Arbiter justification for RW classification:**

The greenhouse effect is fundamentally a radiative-transfer phenomenon operating on planetary scales with an atmospheric column. Expecting it to appear as a conductive property of gas samples reflects a category error in physics that undermines a substantial section of the paper's argument.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
- **Notes:** _____

10.5539_jsd.v17n6p137 — The Scientific Case Against Net Zero: Falsifying the Greenhouse Gas Hypothesis

- **Paper category:** Environmental Science
- **SPOT error category:** Equation / proof (severity tier: errata)
- **SPOT annotation (the one annotated error):** The author used the base-10 logarithm ($\log_{10}$) instead of the natural logarithm ($\ln$, neperian logarithm) in the IPCC radiative forcing equation, resulting in incorrect temperature change calculations.
- **B3 finding (rated RETRACTION-WORTHY, finding #8):**
Location:

The cost-benefit conclusion against Net Zero is asserted without any transparent economic model, discounting assumptions, damage functions, uncertainty analysis, or comparison with adaptation/mitigation benefits.

- **Arbiter justification for RW classification:**

The paper's policy recommendation is presented as a scientific conclusion, but no valid cost-benefit methodology supports it. Economic cost figures are drawn from advocacy reports and combined rhetorically rather than analytically.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
- **Notes:** _____

2401.09543v2 — Token Jumping in Planar Graphs has Linear Sized Kernels

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The proof incorrectly claims a linear kernel size in k , but the correct approach only yields a quadratic kernel, as shown in later work; the original proof thus overstates the achievable bound.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

The final equivalence argument (Claim 7) invokes 'Claim 4' to conclude reconfigurability, but Claim 4 only provides a size bound, not a reconfiguration guarantee. This is a direct logical misapplication at the critical last step of the proof.

- **Arbiter justification for RW classification:**

Both reviewers identified this issue. Reviewer A flagged it as retraction-worthy; Reviewer B noted the incorrect reference and called the equivalence argument insufficient. As written, the final step of the main theorem relies on an inference that does not follow from the cited claim. This is not a typo issue—if the intended claim (e.g., Claim 5) were substituted, its own proof is only a sketch, so the gap remains. The central conclusion is unsupported.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2401.09543v2 — Token Jumping in Planar Graphs has Linear Sized Kernels

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The proof incorrectly claims a linear kernel size in k , but the correct approach only yields a quadratic kernel, as shown in later work; the original proof thus overstates the achievable bound.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

The proof applies the linear-forest reconfiguration argument to $G[J_s \cup J_t \cup J_m]$ without rigorously establishing that this induced subgraph is a linear forest. J_s and J_t are stated to be 'not necessarily independent,' and edges involving boundary vertices are not controlled.

- **Arbiter justification for RW classification:**

Reviewer A identified this as retraction-worthy. The entire equivalence argument depends on the linear-forest structure of this subgraph to apply the reconfiguration result. Without verification that the subgraph has maximum degree ≤ 2 and no cycles, the proof collapses. Reviewer B's concerns about J_s/J_t construction reinforce this: if these sets are not well-characterized, the structural claim about their union is unverifiable.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2401.09543v2 — Token Jumping in Planar Graphs has Linear Sized Kernels

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The proof incorrectly claims a linear kernel size in k , but the correct approach only yields a quadratic kernel, as shown in later work; the original proof thus overstates the achievable bound.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

Claim 7's equivalence proof is incomplete: the induction-based argument does not rigorously verify that each intermediate token move preserves independence, that enough target vertices are available, and that tokens outside the currently discussed class do not create conflicts.

- **Arbiter justification for RW classification:**

Both reviewers flagged this. Reviewer A rated it retraction-worthy; Reviewer B called it a major revision issue but noted the proof is 'incomplete and unconvincing.' Since Claim 7 is the core equivalence statement required for kernelization, and the argument has multiple unverified steps, the main theorem is not established. The severity is at the retraction-worthy/major-revision boundary, but given that this is the proof's central mechanism, the higher rating is appropriate.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2401.16191v2 — From Tripods to Bipods: Reducing the Queue Number of Planar Graphs Costs Just One Leg

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** Lemma 4.2 is incorrect, so the proof relying on it is invalid and requires new technical details to correct the decomposition argument.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

Theorem 4.9 proof is explicitly marked as DRAFT and is incomplete, with undefined notation, inconsistent queue class names, and no verifiable derivation of the 38-queue bound.

- **Arbiter justification for RW classification:**

Theorem 4.9 is the direct basis for the headline result (Theorem 1.5: 38-queue bound for planar graphs). Both reviewers note that the proof is incomplete and the notation is inconsistent. An incomplete draft proof with undefined terms means the central claimed bound is simply not established. This is not a matter of a gap that could be filled—the proof is openly unfinished.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2401.16191v2 — From Tripods to Bipods: Reducing the Queue Number of Planar Graphs Costs Just One Leg

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** Lemma 4.2 is incorrect, so the proof relying on it is invalid and requires new technical details to correct the decomposition argument.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

The degenerate step augmentation (Section 3.2) does not rigorously justify that BFS-layering properties are preserved after adding new vertices and edges, and does not explain how augmented vertices are removed to obtain results for the original graph.

- **Arbiter justification for RW classification:**

Both reviewers flag this independently. Reviewer B notes that adding edges (t_2, v_y) and (t_3, v_y) where t_2, t_3 are on higher layers can violate BFS-layering span requirements. Reviewer A notes the lack of a projection argument back to the original graph. The degenerate step is part of the iterative construction for Theorem 1.2, the core structural result. If augmentation breaks the BFS-layering or if results cannot be transferred back to the original graph, the main theorem is unsupported. However, such augmentation techniques are standard in the graph structure literature and may be justifiable with more careful exposition, so there is some chance this is repairable.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2402.01549v2 — Quantum advantage in zero-error function computation with side information

- **Paper category:** Computer Science
- **SPOT error category:** Equation / proof (severity tier: errata)
- **SPOT annotation (the one annotated error):** The error in Claim 3 is likely due to an incorrect equation or flaw in the logical steps of the proof, which may result from the misuse of mathematical principles, incorrect assumptions, or an invalid derivation that undermines the claim's validity.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

Proposition 1 claims $G \boxtimes H = G \vee H$ (strong product equals OR/disjunctive product) for simple graphs, which is false in general.

- **Arbiter justification for RW classification:**

The strong product and OR (disjunctive) product are well-known to be distinct graph products (see Imrich & Klavžar, 'Product Graphs'). In the strong product, vertices differing in both coordinates are adjacent only if BOTH coordinate pairs are adjacent; in the OR product, adjacency holds if AT LEAST ONE coordinate pair is adjacent. These edge sets differ for most nontrivial graphs. This proposition is foundational: Theorem 3's sandwiching argument, the Lovász theta multiplicativity claims, and the quantum-classical separation all depend on it. Both reviewers independently identified this error. The proof is relegated to a missing appendix, making it unverifiable, but the claim itself is contradicted by standard graph theory. If this equality is used substantively in proofs (not just as a notational convention for a nonstandard definition), the paper's central conclusions collapse.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2402.01549v2 — Quantum advantage in zero-error function computation with side information

- **Paper category:** Computer Science
- **SPOT error category:** Equation / proof (severity tier: errata)
- **SPOT annotation (the one annotated error):** The error in Claim 3 is likely due to an incorrect equation or flaw in the logical steps of the proof, which may result from the misuse of mathematical principles, incorrect assumptions, or an invalid derivation that undermines the claim's validity.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

The converse direction of Lemma 4 incorrectly derives an orthogonal representation from a zero-error quantum protocol by selecting an arbitrary eigenvector from each mixed state.

- **Arbiter justification for RW classification:**

Zero-error distinguishability requires orthogonal supports only for confusable pairs (edges of $G^{\wedge}(m)$), not for all edges of the complement graph as claimed. Furthermore, extracting a single eigenvector from each mixed state does not preserve the pairwise orthogonality structure needed for an orthogonal representation. This gap invalidates the converse of Lemma 4, which is the basis for the quantum rate characterization $R_{\text{quantum}} = \inf_m (1/m) \log \xi(G^{\wedge}(m))$ in Theorem 4. Reviewer A provides detailed technical justification; Reviewer B concurs there is a gap. While there might exist a correct proof via a different argument (e.g., Nayak-type arguments or purification), the proof as written is invalid.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2402.01549v2 — Quantum advantage in zero-error function computation with side information

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- **SPOT error category:** Equation / proof (severity tier: errata)
- **SPOT annotation (the one annotated error):** The error in Claim 3 is likely due to an incorrect equation or flaw in the logical steps of the proof, which may result from the misuse of mathematical principles, incorrect assumptions, or an invalid derivation that undermines the claim's validity.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

Section VI contains invalid graph-complement identities: claims $C_5^{\vee \{v_m\}} = \tilde{C}_5^{\wedge \{v_m\}} = C_5^{\wedge \{v_m\}}$, conflating complement of product with product itself.

- **Arbiter justification for RW classification:**

For self-complementary G , one has $\text{complement}(G \vee H) \cong \tilde{G} \boxtimes \tilde{H} \cong G \boxtimes H$ (using self-complementarity), but this gives $\text{complement}(G^{\vee \{v_m\}}) \cong G^{\wedge \{v_m\}}$, NOT $G^{\vee \{v_m\}} = G^{\wedge \{v_m\}}$. The paper equates a graph with its complement, which for nontrivial graphs is false. The claimed exact value $R_{\text{quantum}}(g) = \frac{1}{2} \log_2 5$ for the C_5 example depends on these identities. Since this is the paper's central showcase of quantum advantage, the quantitative separation is unestablished.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2402.10307v2 — A New Radio to Overcome Critical Link Budgets

- **Paper category:** Computer Science
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the beamforming gain for N antennas is incorrectly calculated as N , when it should actually be N^2 , since coherent beamforming results in the received power increasing by the square of the number of antennas.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

The central claim of equivalent or superior link-budget gain compared to transmit beamforming rests on comparing coherent spatial beamforming gain against a 'temporal combining gain' without proper normalization of total transmitted energy, occupied bandwidth, symbol rate, and noise bandwidth. The two mechanisms are fundamentally different, and the manuscript does not prove their equivalence under identical resource constraints.

- **Arbiter justification for RW classification:**

Both reviewers identify this as the paper's most critical flaw. Beamforming gain arises from coherent spatial superposition at a point in space; the proposed temporal combining gain arises from spreading symbols over more time-frequency dimensions. Without a rigorous proof that these yield the same SNR for the same total radiated power, bandwidth, and data rate, the paper's headline conclusion is unsupported. The numerical examples where $T > N$ and the temporal gain exceeds beamforming gain strongly suggest the comparison is not resource-normalized, making the central conclusion fundamentally broken rather than merely incomplete.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2402.10307v2 — A New Radio to Overcome Critical Link Budgets

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- **SPOT annotation (the one annotated error):** The error is that the beamforming gain for N antennas is incorrectly calculated as N , when it should actually be N^2 , since coherent beamforming results in the received power increasing by the square of the number of antennas.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

Equations (3)–(4) and the claim that a unitary spreading matrix F_b yields a T -fold received-power/SINR improvement are either incorrect or severely under-specified. Matched filtering against orthogonal spreading codes recovers signal components but does not create extra received energy without a corresponding increase in transmitted energy or decrease in data rate.

- **Arbiter justification for RW classification:**

This is the theoretical foundation of the proposed scheme. Reviewer A identifies this as a core error; Reviewer B implicitly flags it by noting the absence of a fundamental limit derivation for temporal combining gain. If the T -fold gain is illusory (i.e., it merely reflects spreading gain that is offset by reduced symbol density), then the entire theoretical framework collapses. There is sufficient concern from both reviews to classify this as retraction-worthy, though some residual uncertainty remains about whether a correct reformulation could rescue partial claims.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
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2402.10307v2 — A New Radio to Overcome Critical Link Budgets

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- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the beamforming gain for N antennas is incorrectly calculated as N , when it should actually be N^2 , since coherent beamforming results in the received power increasing by the square of the number of antennas.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

The numerical example showing temporal combining gain exceeding 12 dB (the 16-antenna beamforming gain) when $T = 18.375 > N = 16$, and claims of 15 dB gain for $T = 46$, $N = 32$, are strong indicators that the gain metric is not physically comparable to beamforming gain and likely double-counts processing gain from reduced symbol density.

- **Arbiter justification for RW classification:**

Both reviewers flag this. Exceeding the coherent array gain in a fair comparison would require consuming additional resources. The manuscript presents this as an advantage without identifying what resource was additionally consumed, which is a hallmark of a flawed normalization. This finding is tightly coupled to the first two and collectively they invalidate the paper's primary conclusion.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2402.16847v2 — The Art of Staying Ahead of Deadlines: Improved Algorithms for the Minimum Tardy Processing Time

- **Paper category:** Computer Science
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the runtime analysis omits the contribution of the processing times within the interval $[d_j - p_j, d_{j+1}]$, which leads to an incomplete and therefore incorrect calculation of the total processing time.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

The dynamic-string representation has internal inconsistencies regarding fixed string length. After splitting at position $d_j - p_j$, the resulting substring has length $d_j - p_j + 1$, not $u = d_{\max} + 1$. Subsequent concat and split operations assume a common universe of size $d_{\max} + 1$ but the actual string lengths do not match, making key algorithmic steps ill-defined.

- **Arbiter justification for RW classification:**

This affects the central algorithmic construction. Without a valid and consistent representation of the indicator strings under split/shift/merge, the core single-machine algorithm is not established. Both reviewers noted this: Reviewer A identified the specific length mismatch, and Reviewer B noted the operations are underspecified to the point of being unverifiable. The issue is fundamental to the paper's main contribution, though a careful rewrite with explicit padding/alignment conventions could potentially repair it.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2402.16847v2 — The Art of Staying Ahead of Deadlines: Improved Algorithms for the Minimum Tardy Processing Time

- **Paper category:** Computer Science
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the runtime analysis omits the contribution of the processing times within the interval $[d_j - p_j, d_{j+1}]$, which leads to an incomplete and therefore incorrect calculation of the total processing time.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

The multi-machine extension ($P_m || \sum p_j U_j$) is not justified. No recurrence for the m -dimensional DP state is given, the 'point-wise trimming' strategy is asserted without proof of correctness, the required multi-dimensional dynamic string data structure is not shown to exist, and the persistence property (Lemma 4.1) is not proved for the multi-machine case.

- **Arbiter justification for RW classification:**

Both reviewers flagged this independently. The m -machine result is a headline contribution but rests on unsupported assertions. Feasibility on m identical machines is not a coordinate-wise independent condition, so 'point-wise' adaptation requires substantial justification that is entirely absent. The multi-dimensional data structure needed is not established in the literature or in this manuscript.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2403.01085v2 — A Strongly Subcubic Combinatorial Algorithm for Triangle Detection with Applications

- **Paper category:** Computer Science
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The analysis in Case 2.1 is faulty because it incorrectly assumes that the triangle detection algorithm will always succeed under certain conditions, but there exist cases where the algorithm fails, making the proof invalid.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

The BFS+ subroutine—the core detection mechanism of the algorithm—is never defined. The manuscript repeatedly references steps and behaviors of BFS+ (e.g., 'See Step 12') but provides no pseudocode or formal specification. Without this, neither correctness nor runtime can be verified.

- **Arbiter justification for RW classification:**

Both reviewers independently identify this as the most critical flaw. The entire paper rests on this subroutine. An algorithm paper whose central algorithm is absent is fundamentally unverifiable. This is not a minor omission—it is the mechanism that supposedly achieves the breakthrough result.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2403.01085v2 — A Strongly Subcubic Combinatorial Algorithm for Triangle Detection with Applications

- **Paper category:** Computer Science
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The analysis in Case 2.1 is faulty because it incorrectly assumes that the triangle detection algorithm will always succeed under certain conditions, but there exist cases where the algorithm fails, making the proof invalid.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

The probabilistic analysis claiming error probability $\leq 10^{-30}$ is unsupported. The sampling calculation ($\exp(-70)$) is mathematically valid in isolation, but it is irrelevant because it only bounds the probability of landing in A, not the probability of detecting a triangle given landing in A. The latter probability is unproved.

- **Arbiter justification for RW classification:**

Both reviewers note that the error bound depends on the unproved detection guarantee. Reviewer B additionally raises concerns about the independence assumptions and parameter regimes. The advertised high-probability guarantee is the basis for the algorithm's claimed correctness, so its failure is fatal to the main theorem.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
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2403.01085v2 — A Strongly Subcubic Combinatorial Algorithm for Triangle Detection with Applications

- **Paper category:** Computer Science
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The analysis in Case 2.1 is faulty because it incorrectly assumes that the triangle detection algorithm will always succeed under certain conditions, but there exist cases where the algorithm fails, making the proof invalid.
- **B3 finding (rated RETRACTION-WORTHY, finding #3):**
Location:

All downstream results—BMM in $O(n^{25/9} \log n)$ (Theorem 3), k -clique in $O(n^{7k/9})$ (Theorem 4), Max-Cut (Theorem 5), and numerous corollaries—depend entirely on the correctness of the triangle detection algorithm (Theorem 2) and therefore collapse if Theorem 2 is invalid.

- **Arbiter justification for RW classification:**

Both reviewers agree that these are pure corollaries of Theorem 2 via standard reductions. None is independently established. Since the base result is broken, the derived claims are unsupported. The paper's headline claims about invalidating the combinatorial k -clique conjecture and related conjectures are therefore unjustified.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2403.18567v2 — The effect of Relativistic Aberration on Cosmological Distances

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the computations incorrectly assume galaxies are in a different reference frame than the observer, leading to fundamentally flawed equations and invalid results throughout the paper.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

Equation (4) identifies cosmological redshift with the pure SR Doppler formula and Eq. (5) uses it to define a global recession velocity for all redshifts, which is then used to parameterize aberration. This identification is not generally valid in FLRW cosmology and is the direct input to the flawed aberration derivation.

- **Arbiter justification for RW classification:**

While one can formally decompose cosmological redshift into chains of infinitesimal Doppler shifts, one cannot extract a single global SR velocity and then apply finite-boost aberration formulas as if source and observer shared a Minkowski chart. This is not a matter of convention but a misapplication of SR in curved spacetime. The entire quantitative framework collapses without this step.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2403.18567v2 — The effect of Relativistic Aberration on Cosmological Distances

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the computations incorrectly assume galaxies are in a different reference frame than the observer, leading to fundamentally flawed equations and invalid results throughout the paper.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

The claim that objects with superluminal recession velocity are unobservable ('we simply can't detect any light signal from the CO') is false in standard FLRW cosmology.

- **Arbiter justification for RW classification:**

Many observed galaxies had or have superluminal recession speeds depending on the definition used, yet are readily observable. This statement reveals a fundamental misunderstanding of the kinematic framework the paper relies on. Since the paper's entire mechanism depends on mapping recession velocity to SR aberration, this misconception propagates into all results.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2403.18567v2 — The effect of Relativistic Aberration on Cosmological Distances

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the computations incorrectly assume galaxies are in a different reference frame than the observer, leading to fundamentally flawed equations and invalid results throughout the paper.
- **B3 finding (rated RETRACTION-WORTHY, finding #3):**
Location:

The revised angular diameter distance definition (Eq. 29, $d_A = r_O(1+z)$) scales oppositely to the standard FLRW result ($d_A = a_0 r / (1+z)$), and appears reverse-engineered to restore Etherington reciprocity after the aberration factor was introduced.

- **Arbiter justification for RW classification:**

The standard Etherington relation $d_L = (1+z)^2 d_A$ is a theorem under broad, well-satisfied conditions. The manuscript first violates it by adding an extra $(1+z)$ to d_L , then redefines d_A ad hoc to restore it. This is not a self-consistent derivation but a post-hoc patch, and the resulting $d_A(z)$ relation is physically wrong.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2403.18567v2 — The effect of Relativistic Aberration on Cosmological Distances

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the computations incorrectly assume galaxies are in a different reference frame than the observer, leading to fundamentally flawed equations and invalid results throughout the paper.
- **B3 finding (rated RETRACTION-WORTHY, finding #4):**
Location:

The CMB argument—that aberration at $z \sim 1000$ would catastrophically smooth anisotropies—contradicts standard treatment and observations. The standard CMB dipole aberration from our peculiar velocity is small and well-measured, not a factor of $\sim 10^6$ smoothing.

- **Arbiter justification for RW classification:**

This is a major downstream conclusion of the paper. It follows from the same invalid global-aberration reasoning applied to the last-scattering surface. The prediction is flatly contradicted by observed CMB anisotropies and their successful modeling in standard cosmology.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
- **Notes:** _____

2403.18567v2 — The effect of Relativistic Aberration on Cosmological Distances

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the computations incorrectly assume galaxies are in a different reference frame than the observer, leading to fundamentally flawed equations and invalid results throughout the paper.
- **B3 finding (rated RETRACTION-WORTHY, finding #5):**
Location:

The derivation of the telescope acceptance-cone transformation (Eqs. 9–13) using SR aberration between distant source and observer frames is not a valid cosmological calculation of collected flux. Photon bundle transport in curved spacetime is governed by null geodesic deviation and optical scalars, not by a global SR solid-angle transformation.

- **Arbiter justification for RW classification:**

This is the specific mathematical step where the extra $(1+z)$ enters. If the solid-angle transformation is invalid (and both reviewers argue it is), then Eqs. 13–16, the revised luminosity distance, and all cosmological consequences are unsupported.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2404.01980v6 — A Simple Ricci Flow Proof of the Uniformization Theorem

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The proof incorrectly assumes that the value of A at $t=0$ and $t=T$ are identical, but without a justification or condition ensuring this, the main inequality in the argument does not necessarily hold.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

The maximum principle comparison in Proposition 3.1 is applied without a valid PDE framework. The isoperimetric ratio I_A is defined as an infimum over curves, and no regularity, classical solvability, or viscosity formulation is established for the evolution equation (2.5). The spatial variable (normal direction r vs. enclosed area A) is left unexplained. Without these, the comparison of the PDE solution with the logistic ODE subsolution is unjustified.

- **Arbiter justification for RW classification:**

This is the main engine of the proof. Both reviewers flag this independently: Reviewer A notes the absence of a fixed spatial domain, boundary conditions, and regularity for the infimum-defined profile; Reviewer B notes the lack of verification that the diffusion and reaction terms permit the claimed ODE comparison. If this comparison fails, the lower bound on I_A^2 is unproven and the entire argument collapses.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2404.01980v6 — A Simple Ricci Flow Proof of the Uniformization Theorem

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The proof incorrectly assumes that the value of A at $t=0$ and $t=T$ are identical, but without a justification or condition ensuring this, the main inequality in the argument does not necessarily hold.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

The proof of Theorem 3.2 contains a flawed contradiction/limiting argument translating the barrier estimate into exponential decay of $\kappa(t)$. Reviewer A identifies that the inequality direction is reversed — the two lower bounds on L_A^2 do not imply the ordering needed for the contradiction. Reviewer B identifies that the constant C depends on A and $C \rightarrow +\infty$ as $A \rightarrow 0$, so $e^{-BT-C} \rightarrow 0$ and the exponential expressions cannot be treated as fixed in the limit, destroying the contradiction.

- **Arbiter justification for RW classification:**

This is the step that converts the isoperimetric estimate into the curvature decay claim, which is the paper's principal conclusion. Both reviewers find independent reasons why this step fails. Whether the issue is inequality direction (Review A) or singular dependence on A (Review B), the result is the same: the claimed estimate $\kappa(t)-1 \leq (\kappa(0)-1)e^{-2t}$ is not established.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2404.04502v2 — The interplay between additive and symmetric large sets and their combinatorial applications

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: errata)
- **SPOT annotation (the one annotated error):** The definition or expression of the operation $(p \odot_t q)$ is incorrect, which undermines the validity of the subsequent results that rely on this operation.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

The claim that $(Z, \odot_{\{l,k\}})$ is an abelian group when $l \mid k(k-1)$ is false. The operation $\odot$ corresponds to multiplication on $\mathbb{Z}+k$ under the isomorphism $x \mapsto lx+k$, and invertibility fails in Z in general (e.g., $l=1, k=1$ gives $x \odot y = xy+x+y$, where $x=1$ has no inverse). This is a foundational algebraic error on which downstream Stone–Čech compactification and ideal arguments depend.

- **Arbiter justification for RW classification:**

The entire framework of transferring ideal structure and idempotent arguments through $(\beta Z, \odot_{\{l,k\}})$ depends on correct identification of the algebraic object. If $(Z, \odot)$ is not a group, the semigroup structure of its Stone–Čech extension is different from what is assumed, and minimal ideals, idempotents, and central sets behave differently. This affects Theorems 4.8, 4.10, 4.11, and Corollary 4.12. However, there is some possibility the authors intend a restricted domain or the error is partially recoverable for semigroup (not group) arguments, hence not absolute certainty.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2404.04502v2 — The interplay between additive and symmetric large sets and their combinatorial applications

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: errata)
- **SPOT annotation (the one annotated error):** The definition or expression of the operation $(p \odot_t q)$ is incorrect, which undermines the validity of the subsequent results that rely on this operation.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

Theorems 4.10 and 4.11 invoke the existence of ultrafilters $p \in \text{cl}(K(\beta Z, +)) \cap E(K(\beta Z, \odot_{\{l,k\}}))$ from earlier theorems that do not establish this existence. Theorem 4.10 cites Theorem 4.8, which does not prove such an intersection is nonempty. Theorem 4.11 cites Theorem 3.3, which concerns only $\odot_t$ (not general $\odot$) and does not explicitly produce idempotents in the intersection. The main partition-regularity conclusions therefore rest on unestablished objects.

- **Arbiter justification for RW classification:**

These are the paper's central results. The entire method depends on finding ultrafilters that simultaneously belong to the closure of the minimal ideal under addition and are idempotent in the $\odot_{\{l,k\}}$ semigroup. Without establishing their existence, the combinatorial conclusions (new partition-regular equations) are unsupported. Both reviewers identify this gap from different angles.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2404.04502v2 — The interplay between additive and symmetric large sets and their combinatorial applications

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: errata)
- **SPOT annotation (the one annotated error):** The definition or expression of the operation $(p \odot_t q)$ is incorrect, which undermines the validity of the subsequent results that rely on this operation.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

The application of Theorem 4.9 in Theorems 4.10 and Corollary 4.12 conflates ultrafilter membership across different semigroup operations. Theorem 4.9 requires an ultrafilter to satisfy two equations in a compatible sense, but the two equations involve different operations (additive vs. $\odot_{\{l,k\}}$). *The paper assumes a single set $A \in p$ simultaneously contains additive structure (arithmetic progressions) and $\odot$ -structure (IP sets) without combinatorial intersection arguments.*

- **Arbiter justification for RW classification:**

This directly undermines the paper's claimed main result and the affirmative answer to Di Nasso's question. Reviewer B identifies this as the critical methodological flaw. However, there is some possibility that the intended argument uses the dual membership of p in both structures more carefully than written, so confidence is slightly lower.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2404.04502v2 — The interplay between additive and symmetric large sets and their combinatorial applications

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: errata)
- **SPOT annotation (the one annotated error):** The definition or expression of the operation $(p \odot_t q)$ is incorrect, which undermines the validity of the subsequent results that rely on this operation.
- **B3 finding (rated RETRACTION-WORTHY, finding #3):**
Location:

Corollary 4.12 states that $x + P(y-x) = z \odot_{\{l,k\}} w$ is *partition regular*, but the proof concludes with $x + P(y-x) = z \oplus w$, which is a different operation. The statement and proof are mismatched.

- **Arbiter justification for RW classification:**

This is the paper's headline result. A mismatch between the operation in the statement and the operation in the proof means the theorem as stated is not proved. If $\oplus_{\{l,k\}}$ was intended, the result is less novel; if $\odot$ was intended, a different proof is needed.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2405.05635v2 — Large Bricks and Join-irreducible torsionfree classes

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** There is a logical gap in the proof of Proposition 3.9, meaning that a necessary step or justification is missing, so the conclusion does not properly follow from the premises given.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

Lemma 3.7 unjustifiably concludes that any nonzero map from a $\lim \rightarrow f$ -simple module F to a simple object S in the HRS heart must be a monomorphism 'by simplicity of F .' The module F has only been shown to be $\lim \rightarrow f$ -simple in the module category, not simple in the heart. No proof or reference bridges these two notions, and Proposition 3.5 (which connects torsionfree almost-torsion modules to simple objects in the heart) applies to a different class of modules. This step is essential for embedding F into a brick B , which is the core of the main theorem.

- **Arbiter justification for RW classification:**

Both reviewers identify this as a critical unsupported implication. Reviewer A calls it retraction-worthy; Reviewer B flags it as major-revision but acknowledges the step is needed for the final construction. The implication is not merely a gap in exposition—it conflates two distinct categorical notions ($\lim \rightarrow f$ -simplicity in $\text{Mod-}\Lambda$ vs. simplicity in the heart), and no known result bridges them automatically. Without this step, the main theorem's proof collapses.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2405.05635v2 — Large Bricks and Join-irreducible torsionfree classes

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** There is a logical gap in the proof of Proposition 3.9, meaning that a necessary step or justification is missing, so the conclusion does not properly follow from the premises given.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

Lemma 3.6 argues that to show every proper quotient of F is torsion, it suffices to check maps from simple modules S into F and examine their cokernels. This tests the wrong direction: maximal proper quotients correspond to surjections $F \twoheadrightarrow S$ onto simples, not injections $S \hookrightarrow F$. The proof as written does not establish that the direct limit F is $\lim_{\rightarrow} f$ -simple.

- **Arbiter justification for RW classification:**

Reviewer A identifies this as retraction-worthy with clear mathematical reasoning. Reviewer B raises concerns about the same lemma from a different angle (factoring maps through direct limits). The directional error is concrete: testing injectivity of maps from simples into F is not equivalent to testing that all proper quotients are torsion. Since Lemma 3.6 is the mechanism producing $\lim_{\rightarrow} f$ -simple objects, its failure breaks the chain to Proposition 3.9. However, there is some possibility the argument could be repaired by reversing the direction, so confidence is slightly lower.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2405.11232v2 — Multiplication formula for Hernandez and Leclerc's quivers with potentials

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that Jacobian algebras do not generally have the Ext-symmetry required for the GLS multiplication formula, and generalized preprojective algebras are not always Jacobian because the nilpotency relations for loops are not derived from the potential.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

Theorem 3.2 (main multiplication formula): The proof imports results from [GLS07] that were established in the preprojective/2-Calabi-Yau setting without verifying the required hypotheses hold for the Jacobian algebra A . Key objects and maps (decomposition Y , map q , notation $E(M1, M2)_Y$) are undefined or insufficiently justified. The critical equation (3.16) is asserted rather than derived. The entire proof is logically incomplete.

- **Arbiter justification for RW classification:**

This is the paper's central technical contribution. Both reviewers independently identified that the proof is incomplete, relies on unjustified transfers from a different algebraic setting, and contains undefined objects. Without a valid proof of this theorem, all downstream results (Propositions 4.3, Theorem 4.5, etc.) are unsupported. The issue is not merely a gap that can be filled with a sentence—it requires verifying that an entire framework applies in a new context.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2405.11232v2 — Multiplication formula for Hernandez and Leclerc's quivers with potentials

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that Jacobian algebras do not generally have the Ext-symmetry required for the GLS multiplication formula, and generalized preprojective algebras are not always Jacobian because the nilpotency relations for loops are not derived from the potential.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

Proposition 4.3 contains a logically invalid contradiction argument: from the multiplication formula it concludes $F_T = 2F_T$, but this requires linear independence of F -polynomials that is never established. The deduction that equality of F -polynomials/ g -vectors implies existence of short exact sequences is also unjustified. Additionally, Ext is computed over the wrong category (script A = cluster algebra, not the module category of A).

- **Arbiter justification for RW classification:**

Both reviewers flag this as a broken argument. The contradiction step is a non-sequitur without proving linear independence of F -polynomials, and the category mismatch (Ext over the cluster algebra vs. the Jacobian algebra) is not merely cosmetic. This proposition is the key link between tensor product simplicity and homological algebra, so its failure undermines a main advertised application.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2405.11232v2 — Multiplication formula for Hernandez and Leclerc's quivers with potentials

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that Jacobian algebras do not generally have the Ext-symmetry required for the GLS multiplication formula, and generalized preprojective algebras are not always Jacobian because the nilpotency relations for loops are not derived from the potential.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

Theorem 4.5 proof is logically broken: it appears to prove Lemma 4.4 rather than Theorem 4.5, with abrupt switches to determinantal modules and i-boxes. Key steps (existence of exact sequences, rigidity implying $T = M_1M_2$ or M_2M_1) are asserted without justification.

- **Arbiter justification for RW classification:**

Reviewer A provides detailed evidence that the proof text does not match the theorem statement and that the argument is spliced from different contexts. This is one of the paper's main advertised applications. While conceivably a drafting error, the mismatch is too severe to be a simple fix—there is no discernible proof of the stated theorem in the manuscript.

- **Human verdict** (mark one):
- ☐ VALID SEVERE — genuine error, SPOT missed it
- ☐ RELATED — real issue but severity overstated
- ☐ FALSE ALARM — not a genuine severe error
- **Notes:** _____

2405.20571v2 — On the principal eigenvalue for compound Poisson processes

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The main flaw is that the proof does not account for a key exception or missing condition, which leads to Theorem 2.1 being stated too generally and thus not holding in all claimed cases.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

Lemma 4.5 is incorrectly proved: the proof conflates conditioning on $\{S_n \in D\}$ with conditioning on the full path survival event $\bigcap_{k=1}^n \{S_k \in D\}$. The conditional distribution of S_n given the entire path stays in D is not uniform on D and depends on path history. The bounded convergence theorem is applied to a ratio whose denominator involves the full survival event, not merely the marginal event $\{S_n \in D\}$. The claimed uniform limit is unsubstantiated.

- **Arbiter justification for RW classification:**

Both reviewers independently identify this as the critical error. Lemma 4.5 is the key technical input converting the jump-chain survival probabilities into the explicit eigenvalue formula of Theorem 2.1. Without it, the entire derivation collapses. The error is not a gap that can be filled by adding a sentence — the conditional distribution given full survival is fundamentally different from the conditional distribution given the marginal event, especially for bounded jumps.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2405.20571v2 — On the principal eigenvalue for compound Poisson processes

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The main flaw is that the proof does not account for a key exception or missing condition, which leads to Theorem 2.1 being stated too generally and thus not holding in all claimed cases.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

Theorem 2.1's explicit formula $\lambda_1^X(D) = (r/|D|) \int_D \int (y-x) dy$, whose principal eigenvalue is not generally equal to the spatial average of this rate. A counterexample structure exists: for domains with weakly connected components, the principal eigenvalue is governed by eigenfunction localization, not the global average. $\int (y-x) dy dx$ is the spatial average of the one-step escape rate, not the principal eigenvalue of the killed generator. The generator of the killed compound Poisson semigroup is a nonlocal integral operator with spatially varying killing rate $r|_{D^c}$

- **Arbiter justification for RW classification:**

Reviewer A provides both the generator calculation and a counterexample argument showing the formula is likely false in general. The averaging over D produces a quantity that does not coincide with the principal eigenvalue of a nontrivial integral operator except in special (e.g., constant-kernel-on- D) cases. This is the paper's central theorem.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
- **Notes:** _____

2405.20571v2 — On the principal eigenvalue for compound Poisson processes

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The main flaw is that the proof does not account for a key exception or missing condition, which leads to Theorem 2.1 being stated too generally and thus not holding in all claimed cases.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

Lemma 4.4's proof contains a scaling error and an incomplete local limit theorem argument. The CLT-based rescaling by $1/\sqrt{n}$ shrinks D to a point, so the approximation $j_n(v/\sigma\sqrt{n}) \approx \gamma(0)$ only shows the density near 0 dominates, not that the conditional distribution on D becomes uniform. The missing $(\sigma\sqrt{n})^{-d}$ normalization factor is dropped without justification. Uniformity in x is claimed but not established.

- **Arbiter justification for RW classification:**

Both reviewers flag this lemma. Reviewer A identifies a fatal scaling error (missing normalization factor); Reviewer B identifies the conceptual flaw that the CLT-based argument does not apply in the large deviations regime relevant to $\{S_n \in D\}$ for fixed bounded D . This lemma feeds directly into Lemma 4.5 and hence Theorem 2.1. While a proper local limit theorem might salvage the statement, the proof as written is fundamentally flawed.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2405.20571v2 — On the principal eigenvalue for compound Poisson processes

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The main flaw is that the proof does not account for a key exception or missing condition, which leads to Theorem 2.1 being stated too generally and thus not holding in all claimed cases.
- **B3 finding (rated RETRACTION-WORTHY, finding #6):**
Location:

Corollary 2.2 (Faber–Krahn inequality: balls uniquely minimize the principal eigenvalue among domains of given volume) depends entirely on Theorem 2.1. Since the theorem's formula is unsupported, the Faber–Krahn result is also unsupported. The rearrangement argument at most proves an inequality for the averaged escape integral, not for the principal eigenvalue.

- **Arbiter justification for RW classification:**

This is a central advertised result of the paper. Its validity is entirely derivative of Theorem 2.1, which is itself unsupported.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2406.03201v2 — Characteristic ideal of the fine Selmer group and results on $\mathbb{C}^\circ$ -invariance under isogeny in the function field case

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error in the proof of Theorem 5.1 involves a flaw or gap in the logical reasoning or calculations that invalidates the original proof, and a corrected version has been provided in arXiv:2407.21431.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

In Part II (Theorem 1.5/6.3 and §6), the Galois group $G = \text{Gal}(K_\infty/K)$ is simultaneously described as an open subgroup of $\text{GL}_2(\mathbb{Z}_\ell)$ and as pro- p , which is incompatible when $\ell \neq p$. This misidentification affects the Iwasawa algebra $\Lambda(G)$, the definition of μ_G , and all transfer arguments.

- **Arbiter justification for RW classification:**

The entire noncommutative Iwasawa-theoretic framework in Part II is built on the structure of G . If G is misidentified, the completed group ring, the $M_H(G)$ condition, and the μ -invariant comparisons are set in the wrong category. This directly invalidates Theorem 1.5/6.3.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2406.04846v2 — Efficient Fault-Tolerant Single Qubit Gate Approximation And Universal Quantum Computation Without Using The Solovay-Kitaev Theorem

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that Equation 6, derived using real arithmetic, is incorrectly applied to a context that uses modulo 2 arithmetic, making the result incompatible with the actual operations on the state ket.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

The manuscript does not solve the stated problem of approximation from a fixed finite gate set. The gate set $\{H, P(\pi\ell/2^m)\}$ depends on ϵ through m , and the set of allowed rotations $P(\pi\ell/2^m)$ for $\ell < 2^m$ contains $O(2^m)$ elements that grow with $1/\epsilon$. This fundamentally changes the problem and makes comparisons to the Solovay-Kitaev theorem and the $\Omega(\log(1/\epsilon))$ lower bound meaningless.

- **Arbiter justification for RW classification:**

Both reviewers identify this independently. The central framing of the paper is as an advance on the finite-gate-set synthesis problem. An ϵ -dependent gate set of exponential size trivializes the problem entirely (one could simply include the target gate). This is not a matter of interpretation — the paper explicitly acknowledges varying the gate set with ϵ but dismisses the issue, which is the core theoretical challenge.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2406.04846v2 — Efficient Fault-Tolerant Single Qubit Gate Approximation And Universal Quantum Computation Without Using The Solovay-Kitaev Theorem

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that Equation 6, derived using real arithmetic, is incorrectly applied to a context that uses modulo 2 arithmetic, making the result incompatible with the actual operations on the state ket.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

The recursive construction for implementing $P(\pi\ell/2^m)$ fault-tolerantly is not properly specified. The recursion direction is ambiguous (doubling vs. halving angles), no explicit base case is given, the referenced circuit figures (Figs. 1 and 2) are missing from the manuscript, and the resource analysis (claimed $O(n)$ overhead) is unsupported with no accounting for state preparation, verification, rejection sampling, or teleportation costs.

- **Arbiter justification for RW classification:**

Reviewer B identifies the mathematical impossibility of the doubling recursion as described; Reviewer A identifies the missing resource analysis. Together, these mean the constructive procedure—the paper's main algorithmic contribution—is both untestable (missing figures) and logically inconsistent (recursion direction). The paper cannot be evaluated or reproduced.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
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2406.04846v2 — Efficient Fault-Tolerant Single Qubit Gate Approximation And Universal Quantum Computation Without Using The Solovay-Kitaev Theorem

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that Equation 6, derived using real arithmetic, is incorrectly applied to a context that uses modulo 2 arithmetic, making the result incompatible with the actual operations on the state ket.
- **B3 finding (rated RETRACTION-WORTHY, finding #3):**
Location:

The final asymptotic complexity statement is internally inconsistent. The conclusion states $O(\epsilon \log(1/\epsilon) \log \log(1/\epsilon) \dots)$ gates, which tends to 0 as $\epsilon \rightarrow 0$ and is nonsensical for a gate count. Earlier sections claim $O(\log(1/\epsilon) \log \log(1/\epsilon) \dots)$. The formula for m in the conclusion also appears dimensionally incorrect.

- **Arbiter justification for RW classification:**

Both reviewers note this inconsistency. While one instance could be a typo, the coexistence of multiple inconsistent formulas for the same quantity (m , gate count, angle precision) across different sections suggests the derivation itself is not controlled. The paper's main quantitative claim—the asymptotic gate count—cannot be determined from the manuscript.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
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2406.15649v2 — Efficient Human Pose Estimation: Leveraging Advanced Techniques with MediaPipe

- **Paper category:** Computer Science
- **SPOT error category:** Statistical reporting (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the angle was calculated incorrectly in Section 3.3, leading to potential inaccuracies in the reported statistical results that rely on this value.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

No actual experimental results are reported. The claimed quantitative improvements (20% IoU, 30% speed reduction) appear only as summary assertions in the text with no supporting tables, figures, confidence intervals, p-values, or any analyzable data.

- **Arbiter justification for RW classification:**

The paper's central conclusions rest entirely on quantitative performance claims, yet no numerical evidence is provided anywhere in the manuscript. This is not a matter of incomplete reporting—the evidence is wholly absent, rendering the conclusions baseless.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
- **Notes:** _____

2406.15649v2 — Efficient Human Pose Estimation: Leveraging Advanced Techniques with MediaPipe

- **Paper category:** Computer Science
- **SPOT error category:** Statistical reporting (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the angle was calculated incorrectly in Section 3.3, leading to potential inaccuracies in the reported statistical results that rely on this value.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

No dataset is identified or described. There are no dataset names, subject counts, number of images/videos, train/test splits, annotation procedures, or acquisition protocols.

- **Arbiter justification for RW classification:**

Without a defined evaluation dataset and ground truth, the reported accuracy and efficiency improvements cannot be interpreted, reproduced, or verified. The empirical foundation of the paper does not exist.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
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2406.15649v2 — Efficient Human Pose Estimation: Leveraging Advanced Techniques with MediaPipe

- **Paper category:** Computer Science
- **SPOT error category:** Statistical reporting (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the angle was calculated incorrectly in Section 3.3, leading to potential inaccuracies in the reported statistical results that rely on this value.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

The proposed methodological contribution is never described. The paper claims 'novel modifications,' 'refined algorithms,' and 'advanced neural network architectures' integrated into MediaPipe but provides no algorithmic details, architectural changes, code modifications, or technical description of what was actually done.

- **Arbiter justification for RW classification:**

A paper claiming methodological innovation must describe the method. Without any concrete description of the purported enhancements, there is nothing to evaluate, reproduce, or connect causally to the claimed performance gains. The scientific contribution is unverifiable.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2406.15649v2 — Efficient Human Pose Estimation: Leveraging Advanced Techniques with MediaPipe

- **Paper category:** Computer Science
- **SPOT error category:** Statistical reporting (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the angle was calculated incorrectly in Section 3.3, leading to potential inaccuracies in the reported statistical results that rely on this value.
- **B3 finding (rated RETRACTION-WORTHY, finding #3):**
Location:

Evaluation metrics are inappropriate or inadequately defined for human pose estimation. IoU and MSE are used without defining what is being compared (keypoints, bounding boxes, skeleton masks), and standard pose estimation metrics (PCK, OKS/mAP, MPJPE) are absent.

- **Arbiter justification for RW classification:**

The 20% IoU improvement headline claim is uninterpretable because it is unclear what IoU measures in this context. If the metric does not correspond to the actual task, the central accuracy claim is meaningless even if numbers were provided.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2406.15649v2 — Efficient Human Pose Estimation: Leveraging Advanced Techniques with MediaPipe

- **Paper category:** Computer Science
- **SPOT error category:** Statistical reporting (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the angle was calculated incorrectly in Section 3.3, leading to potential inaccuracies in the reported statistical results that rely on this value.
- **B3 finding (rated RETRACTION-WORTHY, finding #4):**
Location:

No ground truth measurement procedure is described. It is unclear how reference pose labels were obtained for accuracy evaluation.

- **Arbiter justification for RW classification:**

Accuracy metrics require reference labels. Without knowing the source and quality of ground truth, reported accuracy improvements are unvalidatable. Combined with the absence of dataset details, this confirms the empirical claims are unsupported.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2406.15649v2 — Efficient Human Pose Estimation: Leveraging Advanced Techniques with MediaPipe

- **Paper category:** Computer Science
- **SPOT error category:** Statistical reporting (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the angle was calculated incorrectly in Section 3.3, leading to potential inaccuracies in the reported statistical results that rely on this value.
- **B3 finding (rated RETRACTION-WORTHY, finding #5):**
Location:

Computational efficiency claims (30% processing time reduction) are reported without hardware specifications, software environment, image resolution, latency measurement protocol, or any experimental details.

- **Arbiter justification for RW classification:**

Runtime performance is entirely dependent on measurement conditions. Without these details, the efficiency claim cannot be interpreted or reproduced and is therefore unsupported as presented.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
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2407.02110v2 — On the uniqueness of the strictly convex quadrilateral central configuration with a fixed angle

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** Lemma 1 contains incorrect equations or logical steps, undermining the validity of the proof and making the results that rely on it, including Proposition 2 and the main theorem, unreliable.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

The proof works on the enlarged set M^+ (defined by $l=l_0$, $C=0$, $r_{ij}>0$) rather than the actual configuration space of realizable strictly convex quadrilaterals. M^+ contains non-realizable distance vectors (violating triangle inequalities, $F_4<0$, 3-body-collinear configurations). Uniqueness of critical points on M^+ does not imply uniqueness on the physical configuration set unless all critical points on M^+ are shown to be realizable, which is never established. Additionally, 3-body-collinear configurations in M^+ are dismissed as non-central-configurations but are never excluded as critical points of $U|_{M^+}$, invalidating the Morse-theoretic count.

- **Arbiter justification for RW classification:**

This is the fundamental logical gap identified by Reviewer A (findings 1, 6, 10). The main theorem's conclusion is drawn from a critical-point count on a domain that is strictly larger than the relevant configuration space. Without proving that every critical point on M^+ corresponds to a realizable convex quadrilateral, the uniqueness claim for actual central configurations is unsupported. This is not a matter of missing detail — it is a structural flaw in the proof architecture.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.02110v2 — On the uniqueness of the strictly convex quadrilateral central configuration with a fixed angle

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** Lemma 1 contains incorrect equations or logical steps, undermining the validity of the proof and making the results that rely on it, including Proposition 2 and the main theorem, unreliable.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

The constraint $C(r) = 2r_{12}r_{34}\cos\theta + r_{23}^2 + r_{14}^2 - r_{24}^2 - r_{13}^2 = 0$ involves the angle θ between opposite sides, which is itself a function of the mutual distances r . The manuscript treats $\cos\theta$ as if it were a fixed external parameter throughout the Morse-theoretic analysis and the Euler characteristic computation (Lemma 3). This means M^+ is not a well-defined fixed submanifold of $\mathbb{R}^6$ — it changes with the point r . The topological analysis (contractibility, fibration structure, $\chi=1$) is performed on an object that is not properly defined.

- **Arbiter justification for RW classification:**

Reviewer B's central finding. If θ is not a fixed parameter but depends on r , then M^+ as defined is not a standard level set of a smooth function, and the entire Morse-theoretic framework (which requires a fixed constraint manifold) breaks down. This interacts with and compounds the domain issue above. There is some possibility that the authors intend to fix θ as a parameter and study the family of central configurations at each θ , but even under that charitable reading the arguments are not valid because the constraint C depends on θ in a way that was not properly handled.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.02110v2 — On the uniqueness of the strictly convex quadrilateral central configuration with a fixed angle

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** Lemma 1 contains incorrect equations or logical steps, undermining the validity of the proof and making the results that rely on it, including Proposition 2 and the main theorem, unreliable.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

Lemma 3 ($\chi(M^*)=1$) is inadequately proven. The arguments invoke 'fibrations' from projections with contractible fibers but never establish local triviality or bundle structure. Contractible fibers over a contractible base do not in general yield a contractible total space. The analysis contains inconsistent variable/domain descriptions, treats the $\cos\theta=0$ case incompletely, and the passage from equal masses to arbitrary masses via 'continuous deformation invariance of χ ' is asserted without constructing an explicit homeomorphism or showing that M^* varies continuously in a topology-preserving manner as masses change.

- **Arbiter justification for RW classification:**

Both reviewers flag this (A-finding 4, A-finding 5, B-findings 4-5). The Euler characteristic is the linchpin of the uniqueness argument: $\alpha_0 = \chi(M^*) = 1$ forces exactly one minimum. If $\chi(M^*)=1$ is not established, the entire Morse counting argument collapses. The topological claims are hand-wavy and miss standard requirements for the tools invoked. Since this is not an auxiliary lemma but a load-bearing step, its failure is fatal to the main theorem.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.02110v2 — On the uniqueness of the strictly convex quadrilateral central configuration with a fixed angle

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** Lemma 1 contains incorrect equations or logical steps, undermining the validity of the proof and making the results that rely on it, including Proposition 2 and the main theorem, unreliable.
- **B3 finding (rated RETRACTION-WORTHY, finding #3):**
Location:

Lemma 4's Morse-theoretic argument requires M^* to be a compact smooth manifold (or a manifold with boundary where boundary behavior is carefully handled). M^* has positivity constraints ($r_{ij} > 0$) creating boundary/corner structure. No proof is given that $U_{\lfloor} \{M^*\}$ is a proper Morse function, that boundary critical points are excluded, or that standard Morse equalities apply to this domain.

- **Arbiter justification for RW classification:**

Reviewer A's finding 2, partially echoed by Reviewer B's finding 6 on boundary behavior. The Morse equality $\sum (-1)^q \alpha_q = \chi$ requires specific conditions on the domain and the function. The manuscript applies it without verification. Combined with the issues above, this represents another independent failure of the proof's central mechanism. Downgrading to MAJOR-REVISION might be warranted if the other issues were absent, but in context, this compounds the fundamental breakdown.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.04410v3 — On the Existence of an Extremal Function for the Delsarte Extremal Problem

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: errata)
- **SPOT annotation (the one annotated error):** The error is that the uniform convergence on compact subsets cannot be concluded without first establishing that the limit function f is continuous and satisfies $f(0)=1$, which has not been shown at this point in the proof.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

The proof of Theorem 8 asserts that the admissible class $G_G(\Omega)$ is contained in a closed, bounded subset of $L^2(G)$ and hence is weakly sequentially compact. No argument is given that admissible functions belong to $L^2(G)$ or have uniformly bounded L^2 -norms. The pointwise bound $|f| \leq 1$ and finite measure of Ω only control the positive part; the negative part may live on a set of infinite measure, so $\int |f|^2$ can be infinite. Without L^2 -boundedness, the weak compactness extraction that produces the candidate extremizer is invalid, and the entire existence proof collapses. Theorem 9 depends on Theorem 8, so the main result is unsupported.

- **Arbiter justification for RW classification:**

Both reviewers flagged this independently with high confidence. The gap is not a matter of missing details—it concerns a claim that appears to be false under the paper's hypotheses. If the functions are not in $L^2(G)$, weak compactness in L^2 cannot be invoked, and no alternative compactness framework is provided. The entire existence proof rests on this step.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.06913v2 — A Simple, Nearly-Optimal Algorithm for Differentially Private All-Pairs Shortest Distances

- **Paper category:** Computer Science
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is the incorrect assumption in Sections 3 and 4 that the topology of the shortest path trees is public, which may not be true; additionally, in Section 3, Lemma 2.4 is applied inappropriately under this false assumption.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

Lemma 3.2 (tree release primitive) has an incorrect sensitivity analysis. The paper claims the released vector of distances has ℓ_1 -sensitivity 1 because subtrees are disjoint, but changing one edge weight can affect multiple distances whose shortest paths traverse that edge, yielding sensitivity larger than 1. The noise calibration is therefore insufficient for the claimed privacy guarantee.

- **Arbiter justification for RW classification:**

This primitive is invoked repeatedly throughout Sections 3 and 4 as the building block for all main algorithms. If the privacy proof of the primitive is invalid, every downstream theorem inherits the flaw. Reviewer A identifies this precisely; Reviewer B's 'composition of compositions' concern is a related but differently framed version of the same core issue (the per-tree privacy/accuracy tradeoff is not as claimed). Because the paper's central contribution is the family of DP algorithms built on this primitive, this undermines the main conclusions.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.06913v2 — A Simple, Nearly-Optimal Algorithm for Differentially Private All-Pairs Shortest Distances

- **Paper category:** Computer Science
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is the incorrect assumption in Sections 3 and 4 that the topology of the shortest path trees is public, which may not be true; additionally, in Section 3, Lemma 2.4 is applied inappropriately under this false assumption.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

Advanced composition is misapplied in the approximate- (ϵ, δ) -DP results. The manuscript uses per-mechanism privacy budget scaling as $1/k$ (or $1/s$) rather than $1/\sqrt{k}$ (or $1/\sqrt{s}$), inconsistent with the paper's own stated Lemma 2.2. This affects the error bounds in Theorems 3.2, 4.2, and related results.

- **Arbiter justification for RW classification:**

This is not a constant-factor issue; it changes the asymptotic dependence on s (and hence n) in the final error rate. The headline $\tilde{O}(n^{1/4}/\epsilon)$ bound for approximate DP is derived from a balancing argument that relies on this incorrect scaling. If corrected, the optimal parameter choice and resulting rate change materially. Both reviewers flag this: Reviewer A pinpoints the specific algebraic discrepancy ($1/s$ vs $1/\sqrt{s}$), and Reviewer B identifies the balancing argument as fundamentally flawed for related reasons.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.06913v2 — A Simple, Nearly-Optimal Algorithm for Differentially Private All-Pairs Shortest Distances

- **Paper category:** Computer Science
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is the incorrect assumption in Sections 3 and 4 that the topology of the shortest path trees is public, which may not be true; additionally, in Section 3, Lemma 2.4 is applied inappropriately under this false assumption.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

Section 4.2, Lemma 4.4: $|A_i|$ is modeled as $\text{Binomial}(n, p_i)$, but A_i is a union of fixed-size subsets sampled without replacement, so the vertex indicators are dependent and the binomial model is invalid.

- **Arbiter justification for RW classification:**

This lower bound on $|A_i|$ feeds into Lemma 4.5 and the subsequent stretch analysis for the Thorup-Zwick-style oracle. The probabilistic foundation for the $2k-1$ stretch guarantee is therefore unsupported. Reviewer A identifies this clearly as retraction-worthy; Reviewer B flags it as a major concern though with slightly less certainty. The dependence issue is real and not easily hand-waved away, though it might be repairable with a more careful analysis (e.g., negative association arguments). Given that no such repair is present, the current proof is invalid.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.18406v2 — A form of refined Roth's theorem and its application to the \$abc\$-conjecture

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** Theorem 1.6 contains an error because one or more of the equations or logical steps used in the proof are incorrect, possibly due to an invalid assumption or a misapplied result that invalidates the theorem's conclusion.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

The derivation of the central inequality (14) in Theorem 2.1 relies on a local-to-global estimate that conflates valuations across places, mishandles normalizations between the base field k and extension field K (suppressing $[K_w:k_v]$ factors), and uses undefined or inconsistent notation $(N_{\{x'\}}(x), S_{\{j,x'\}}(x), 2m \lceil x'Q(x), \infty \rceil)$. The theorem statement and proof are not mathematically well-posed.

- **Arbiter justification for RW classification:**

Both reviewers flag this as the foundation of all subsequent results. If the central inequality is not coherently stated or proved, the entire proof chain collapses. The notation issues are not cosmetic—they prevent verification of the mathematical content.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.18406v2 — A form of refined Roth's theorem and its application to the \$abc\$-conjecture

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** Theorem 1.6 contains an error because one or more of the equations or logical steps used in the proof are incorrect, possibly due to an invalid assumption or a misapplied result that invalidates the theorem's conclusion.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

The transition from inequality (15) to (16) and the subsequent derivation of (17)–(18) are algebraically unjustified. The dependence on the index j is not handled consistently, sign conventions are confused between proximity and valence functions, and the place-by-place estimate (17) mixes $|\cdot|_v$ and $|\cdot|(x) \leq 2m(x'Q(x), \infty) + O(1)$ therefore lacks a valid foundation.} with dimensionally inconsistent constants c_v . The conclusion $S_{\{x'$

- **Arbiter justification for RW classification:**

This is the mechanism by which a sum of proximity terms is replaced by a single proximity term — the key technical trick. Both reviewers identify independent errors here. Without this step, inequality (19) and hence Theorem 2.1 do not follow.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.18406v2 — A form of refined Roth's theorem and its application to the \$abc\$-conjecture

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** Theorem 1.6 contains an error because one or more of the equations or logical steps used in the proof are incorrect, possibly due to an invalid assumption or a misapplied result that invalidates the theorem's conclusion.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

The proof of Theorem 1.4 claims that $y = 1/(x'Q(x))$ is an algebraic integer for algebraic integer x . This integrality claim is not established: the construction of x' via ideals a_0 and a_∞ is incomplete, and the divisibility $x' \mid \prod_j (x - a_j)$ in the ring of integers is never verified. Without integrality, the crucial conclusion $m(x'Q(x), \infty) = O(1)$ does not follow.

- **Arbiter justification for RW classification:**

Both reviewers identify this as essential to the proof of Theorem 1.4. The integrality claim is what eliminates the extra error term and yields the main result. Reviewer A provides a detailed explanation of why the claim is unproved; Reviewer B notes the circularity when attempting to extend to all algebraic numbers.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.18406v2 — A form of refined Roth's theorem and its application to the \$abc\$-conjecture

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** Theorem 1.6 contains an error because one or more of the equations or logical steps used in the proof are incorrect, possibly due to an invalid assumption or a misapplied result that invalidates the theorem's conclusion.
- **B3 finding (rated RETRACTION-WORTHY, finding #3):**
Location:

The application of Dirichlet's unit theorem to conclude $m(y, 0) = O(1)$ is invalid as presented. The claimed uniform comparability $|y|_v \leq c(\kappa)|y|_w$ for arbitrary algebraic integers y and Archimedean places v, w does not hold without unit normalization. Even with such normalization, the bound would depend on y , not just on κ . This is the final step in proving Theorem 1.4.

- **Arbiter justification for RW classification:**

Both reviewers flag this independently. This is the concluding step that converts auxiliary estimates into the main theorem. Without a valid argument here, the proof of Theorem 1.4 fails at its conclusion.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.18406v2 — A form of refined Roth's theorem and its application to the \$abc\$-conjecture

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** Theorem 1.6 contains an error because one or more of the equations or logical steps used in the proof are incorrect, possibly due to an invalid assumption or a misapplied result that invalidates the theorem's conclusion.
- **B3 finding (rated RETRACTION-WORTHY, finding #4):**
Location:

Theorem 1.5 uses the undefined notion of 'simple number' and claims, for $q = 2$, that $y = 1/(x'Q(x))$ is an algebraic integer by the same (already unproved) argument. The definition is not shown to be preserved under the relevant transformations, making Theorem 1.5 unverifiable.

- **Arbiter justification for RW classification:**

Theorem 1.5 is the bridge to the abc-conjecture corollary (Theorem 1.6). Both reviewers note the undefined terminology and the reliance on the same flawed integrality argument.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.19061v3 — A Relationship Between Nonphysical Quasi-probabilities and Nonlocality Objectivity

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the calculation incorrectly claims different eigenvalues for $\beta^T \beta$ in the right-identity case, when in fact the eigenvalues remain unchanged for both the left- and right-identity cases, invalidating the argument's main result.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

The Horodecki CHSH nonlocality criterion is invariant under party exchange ($A \leftrightarrow B$ swap), making 'swap-intolerant nonlocality' impossible within the standard bipartite formalism. Swapping parties sends $T \rightarrow T^T$, but the eigenvalues of $T^T T$ are identical to those of $T T^T$, so the Horodecki quantity M is unchanged. The paper's central concept is therefore mathematically incoherent.

- **Arbiter justification for RW classification:**

This is a theorem-level impossibility result: no 2-qubit operator (positive or not) can have its CHSH nonlocality status change under party relabelling when assessed via the Horodecki criterion. The paper's main theorem directly contradicts this mathematical fact. This is not a matter of interpretation or methodology—it is a fundamental error that renders the central conclusion unsupportable.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.19061v3 — A Relationship Between Nonphysical Quasi-probabilities and Nonlocality Objectivity

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the calculation incorrectly claims different eigenvalues for $\beta^T \beta$ in the right-identity case, when in fact the eigenvalues remain unchanged for both the left- and right-identity cases, invalidating the argument's main result.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

The Horodecki criterion is derived and proven only for valid quantum states (positive semidefinite, trace 1). The manuscript applies this criterion to operators with negative eigenvalues without providing any justification that the criterion retains meaning for non-physical operators. The concept of 'nonlocality' (violation of local hidden variable models via Bell-CHSH inequality) is undefined for non-PSD matrices because measurement probabilities can become negative.

- **Arbiter justification for RW classification:**

Both reviewers flag this issue. Even setting aside the swap-invariance problem, the paper's entire framework—extending the Horodecki nonlocality criterion to non-physical operators and interpreting $M > 1$ as 'nonlocality'—lacks any mathematical or physical foundation. The Horodecki theorem's proof relies on properties specific to quantum states. Applying its formula to non-states and drawing conclusions about nonlocality is a category error. Combined with the swap-invariance issue, this further confirms the paper's conclusions are fundamentally unsupported.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
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2407.19061v3 — A Relationship Between Nonphysical Quasi-probabilities and Nonlocality Objectivity

- **Paper category:** Physics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that the calculation incorrectly claims different eigenvalues for $\beta^{\{T\}}\beta$ in the right-identity case, when in fact the eigenvalues remain unchanged for both the left- and right-identity cases, invalidating the argument's main result.
- **B3 finding (rated RETRACTION-WORTHY, finding #3):**
Location:

The existence proof in Theorem 1.4 (construction of SIN states via rotation into the region described by Eq. (9)) is unsupported because it relies on both the erroneous Eq. (8)/(9) target region and on an unjustified claim that the relevant properties are preserved under the prescribed rotation.

- **Arbiter justification for RW classification:**

The constructive heart of the paper's theorem depends on a target region derived from an algebraic error. Since the target region does not actually differ between original and swapped states (per the swap-invariance argument), there are no SIN states to construct. The existence proof is void.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
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2407.19323v3 — MSP-MVS: Multi-granularity Segmentation Prior Guided Multi-View Stereo

- **Paper category:** Computer Science
- **SPOT error category:** Experiment setup (severity tier: errata)
- **SPOT annotation (the one annotated error):** The problem is that there is a significant flaw in how the experiment was designed in the Multi-granularity section, which affects the reliability of the resulting data analysis and the validity of the study's conclusions.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

The manuscript presents two fundamentally different and incompatible methods (MSP-MVS and TSAR-MVS) as a single contribution. MSP-MVS uses Semantic-SAM for multi-granularity segmentation prior, anchor equidistribution, and iterative local search, while TSAR-MVS uses Roberts edge detection, Hough line detection, confidence-based outlier filtering, superpixel RANSAC refinement, and textureless-aware segmentation. The title, abstract, and conclusion refer to MSP-MVS, but large central sections describe TSAR-MVS. No explanation of the relationship between the two methods is provided.

- **Arbiter justification for RW classification:**

Both reviewers independently identified this as the central fatal flaw. The experimental results cannot be attributed to a single, well-defined method. The reader cannot determine which algorithm was actually implemented and evaluated. This is not an editorial oversight but a structural inconsistency that breaks the fundamental scientific contract: the method must be uniquely defined for results to be meaningful. All benchmark claims, state-of-the-art assertions, and generalization conclusions are rendered unsupported because the evaluated system is undefined.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2407.19323v3 — MSP-MVS: Multi-granularity Segmentation Prior Guided Multi-View Stereo

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- **SPOT error category:** Experiment setup (severity tier: errata)
- **SPOT annotation (the one annotated error):** The problem is that there is a significant flaw in how the experiment was designed in the Multi-granularity section, which affects the reliability of the resulting data analysis and the validity of the study's conclusions.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

The ablation studies describe components (e.g., 'edge correction (w/o. CRF)', 'aggregation (w/o. Agr.)', 'sector averaging', 'anchor clustering') that are not defined in the main methods section and do not clearly correspond to either the MSP-MVS or TSAR-MVS pipeline descriptions. The ablations cannot be mapped to a specific, coherent method.

- **Arbiter justification for RW classification:**

Ablation studies are the primary evidence for the contribution of individual components. When the ablated components cannot be matched to the described method—and the method itself is ambiguous between two different systems—the ablation evidence is meaningless. This compounds the core inconsistency and independently undermines the paper's claims about which components drive performance. Both reviewers flagged this, though Reviewer A classified it as retraction-worthy and Reviewer B as major-revision. Given the direct coupling to the identity confusion, retraction-worthy is appropriate.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
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2408.16162v2 — Every Polish group has a non-trivial topological group automorphism

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that extending a non-trivial automorphism of L by fixing elements of a maximal independent subset Y of U can inadvertently redefine the automorphism on elements of L that are generated by Y , making the extension invalid.
- **B3 finding (rated RETRACTION-WORTHY, finding #0):**
Location:

Lemma 2.3: The Zorn's lemma argument is invalid. The union $U_I = \bigcup_{\alpha} U_{\alpha}$ of a chain of identity neighborhoods is not shown to be an identity neighborhood (i.e., an element of the poset). Neighborhoods of the identity in a topological group must satisfy specific properties that arbitrary unions need not preserve. Without verifying the poset is inductive, the maximal element U is not obtained.

- **Arbiter justification for RW classification:**

Reviewer A identifies a concrete mathematical error: the upper bound construction fails because the union of identity neighborhoods need not be an identity neighborhood. This is not a gap that can be trivially filled—it is a fundamental flaw in the Zorn's lemma application. Lemma 2.3 is a critical building block for the main theorem. Reviewer B also flags problems with Lemma 2.3 but rates it MAJOR-REVISION; however, Reviewer A's more detailed analysis is persuasive that the error is structural rather than merely a matter of missing justification.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2408.16162v2 — Every Polish group has a non-trivial topological group automorphism

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that extending a non-trivial automorphism of L by fixing elements of a maximal independent subset Y of U can inadvertently redefine the automorphism on elements of L that are generated by Y , making the extension invalid.
- **B3 finding (rated RETRACTION-WORTHY, finding #1):**
Location:

Lemma 2.3 contradiction step: $W := U \cup V'$ is claimed to be an identity neighborhood in the poset, but V' is constructed as a nonempty open subset of V , not necessarily containing 0_G . Thus W need not be an identity neighborhood, and the maximality contradiction collapses.

- **Arbiter justification for RW classification:**

This is a distinct error within the same lemma. The contradiction that would establish property (2) depends on W being in the poset, which requires W to be a neighborhood of 0 . If V' does not contain 0 , this fails. Reviewer A's analysis is specific and convincing.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2408.16162v2 — Every Polish group has a non-trivial topological group automorphism

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that extending a non-trivial automorphism of L by fixing elements of a maximal independent subset Y of U can inadvertently redefine the automorphism on elements of L that are generated by Y , making the extension invalid.
- **B3 finding (rated RETRACTION-WORTHY, finding #2):**
Location:

Lemma 2.3 contains incorrect or incoherent set-theoretic/group-theoretic computations: sums and intersections are conflated, and the claimed disjointness $(x+V') \cap (y+V') = \emptyset$ is not correctly derived from the preceding expressions involving $x+x+U_x$ and $y+y+U_y$ in a Boolean group.

- **Arbiter justification for RW classification:**

Reviewer A identifies that the displayed computation is internally inconsistent. While in a Boolean group $x+x=0$, so $x+x+U_x = U_x$, the leap to the claimed disjointness of translated sets is not logically justified. This compounds the other errors in Lemma 2.3. Confidence is slightly lower because some of this might be a notational issue, but the mathematical content as written is incorrect.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
-

2408.16162v2 — Every Polish group has a non-trivial topological group automorphism

- **Paper category:** Mathematics
- **SPOT error category:** Equation / proof (severity tier: retract)
- **SPOT annotation (the one annotated error):** The error is that extending a non-trivial automorphism of L by fixing elements of a maximal independent subset Y of U can inadvertently redefine the automorphism on elements of L that are generated by Y , making the extension invalid.
- **B3 finding (rated RETRACTION-WORTHY, finding #3):**
Location:

Proof of Theorem 1.1: The inference that H is dense in G from item (2) of Lemma 2.3 is incorrect. Item (2) states $G \setminus (L+U)$ has empty interior, which means $L+U$ is dense, but the proof conflates $L+U$ with $U + \text{span}\{x_i\}$ and does not correctly establish that H contains a dense subset.

- **Arbiter justification for RW classification:**

Density of H is essential for applying the extension lemma (Lemma 2.2). Reviewer A correctly notes that the logical chain from item (2) to density of H is broken. Without density, the automorphism cannot be extended to all of G , and the main theorem fails.

- **Human verdict** (mark one):
 - ☐ VALID SEVERE — genuine error, SPOT missed it
 - ☐ RELATED — real issue but severity overstated
 - ☐ FALSE ALARM — not a genuine severe error
 - **Notes:** _____
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